TITLE: IMPORT DATA USING TRANSFORM MAPS (SPREADSHEET)

Project Description:

This project focuses on automating the ingestion, validation, and mapping of bulk data from external spreadsheets (Excel/CSV) into targeted ServiceNow tables using **Import Sets** and **Transform Maps**. By leveraging automated field mapping and custom transform scripts, the system processes raw external data seamlessly into structured database records while preserving data integrity and avoiding duplicate entries.

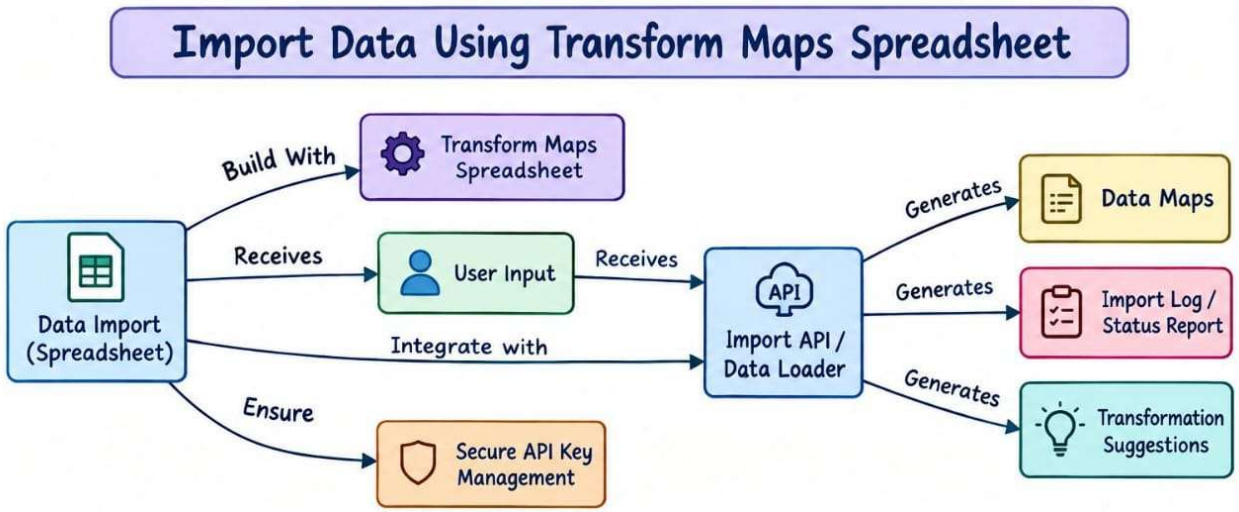
Scenarios

Scenario 1: An HR administrator uploads a spreadsheet containing 500 new employee records into the Import Set Table. The Transform Map matches "Employee ID" as a coalesce key, updating existing employee profiles and inserting missing ones automatically.

Scenario 2: An IT Asset Manager imports legacy hardware asset data from Excel. The system uses field maps to transform raw date formats and reference fields (e.g., Department names) into valid ServiceNow references while logging any mismatched data.

Scenario 3: A Customer Service team bulk-imports incident tickets from a third-party vendor file. Scripted transform logic automatically sets priority levels based on urgency metrics present in the uploaded spreadsheet.

Technical Architecture:



- Frontend / User Interface: ServiceNow Data Import Module & Import Set Web Interface.
- Staging Layer (Import Set Table): Temporary holding table created dynamically to store raw spreadsheet rows before transformation.
- Transformation Engine: ServiceNow Transform Map engine equipped with Field Maps, Coalesce Rules, and Transform Scripts (OnBefore, OnAfter).
- Target Layer: ServiceNow System Tables (e.g., sys_user, alm_hardware, incident).

Pre-requisites:

- ServiceNow System Administration Knowledge: Familiarity with ServiceNow Navigation, Tables, and Data Schema.

- Spreadsheet Handling Skills: Data cleaning, CSV/XLSX formatting, and structured header alignment.

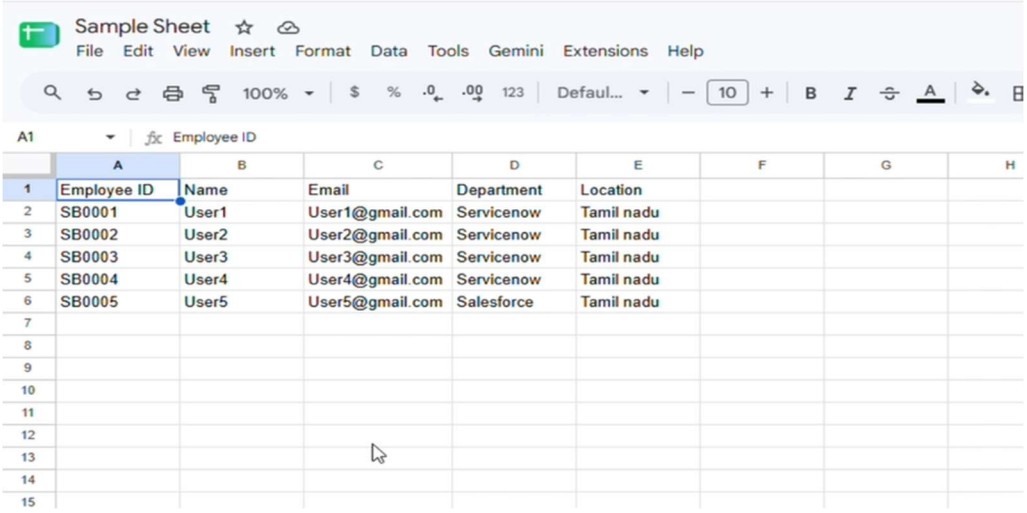
- JavaScript Basics: Scripting knowledge for writing custom ServiceNow Transform Map scripts.
- Data Modeling: Understanding of primary keys, unique identifiers (Coalesce), and relational database fields.

Project Workflow:

Milestone 1: Creation Of Spreadsheet And Table

Activity 1: Creation of Spreadsheet

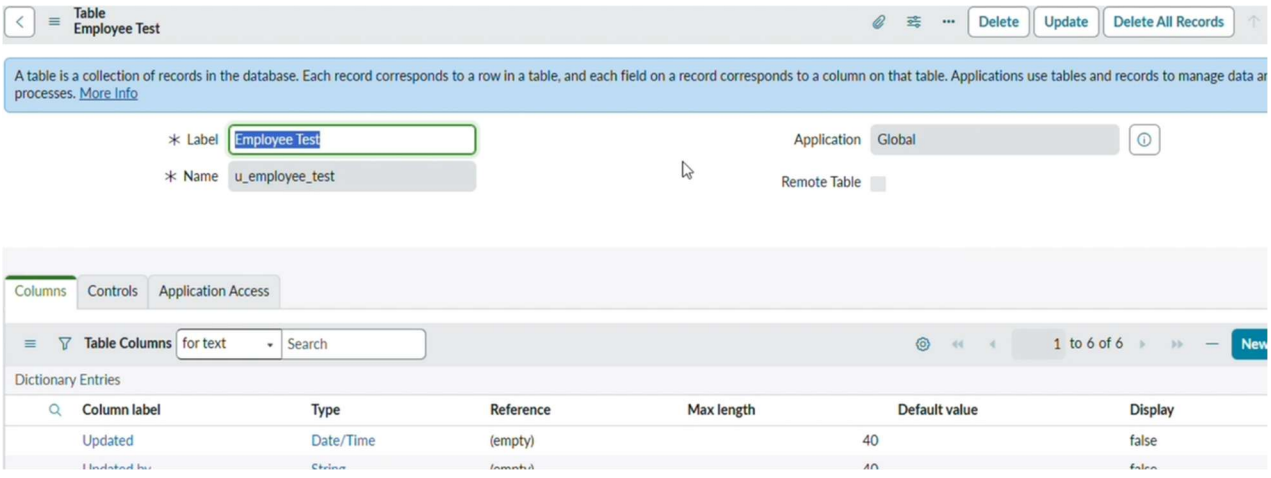
- Open New Google Spreadsheet
- Create New Google Spreadsheet with sample data as shown below
- Download the spreadsheet as Sample Spreadsheet.xlsx format.

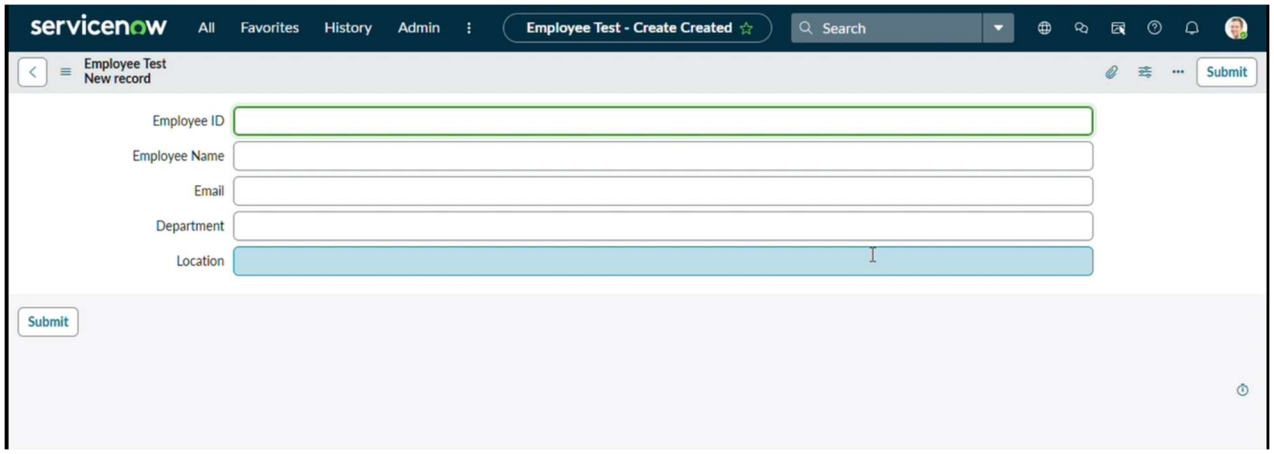


	A	B	C	D	E
1	Employee ID	Name	Email	Department	Location
2	SB0001	User1	User1@gmail.com	Servicenow	Tamil nadu
3	SB0002	User2	User2@gmail.com	Servicenow	Tamil nadu
4	SB0003	User3	User3@gmail.com	Servicenow	Tamil nadu
5	SB0004	User4	User4@gmail.com	Servicenow	Tamil nadu
6	SB0005	User5	User5@gmail.com	Salesforce	Tamil nadu

Activity 2: Creation of Tables

- Tables > Create New
- Label: Employee Test Name: u_employee_test
- After creating and saving the Employee Test table
- scroll down and go to related links
- click on show form
- On the form, locate the **Form Context Menu / Additional Actions** option on the left side of the form header.\
- Click the **Form Context Menu / Additional Actions** icon.
- Click configure , select **Form Layout**.
- The **Form Layout** page will open, where you can configure the fields displayed on the form
- After opening **Form Layout**, click **New** to create a new field.
- Enter the field details:
- **Column Label:** Employee ID
- **Type:** String
- Click **Add** to create the field.
- Repeat the same steps to create the following fields:
- **Employee Name** – Type: String
- **Email** – Type: String
- **Department** – Type: String
- **Location** – Type: String
- After creating all the required fields, click **Save**.
- Verify that the newly created fields are displayed on the **Employee Test** form
- Click on Submit or Save

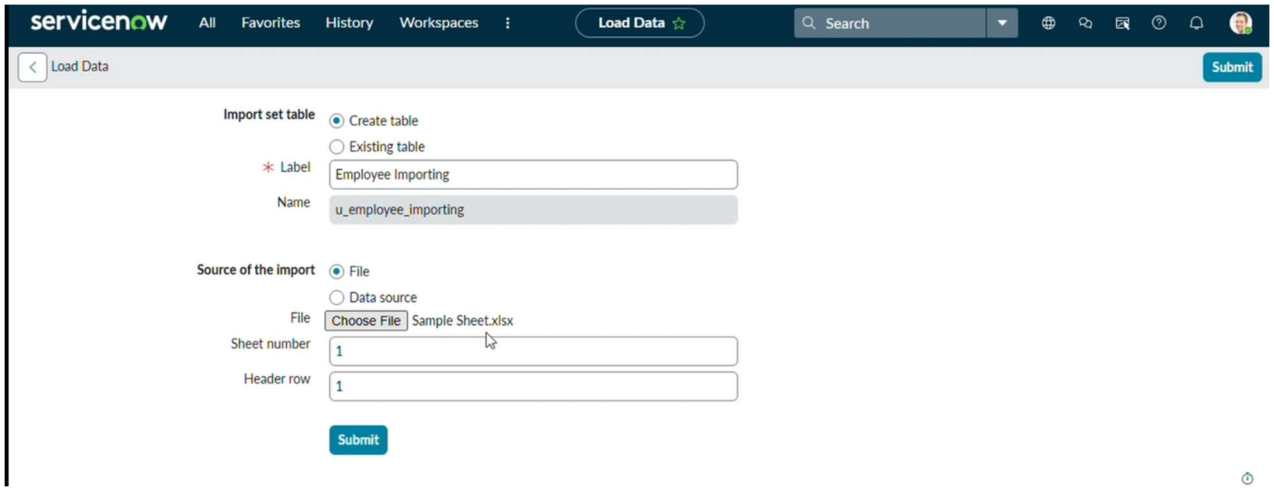


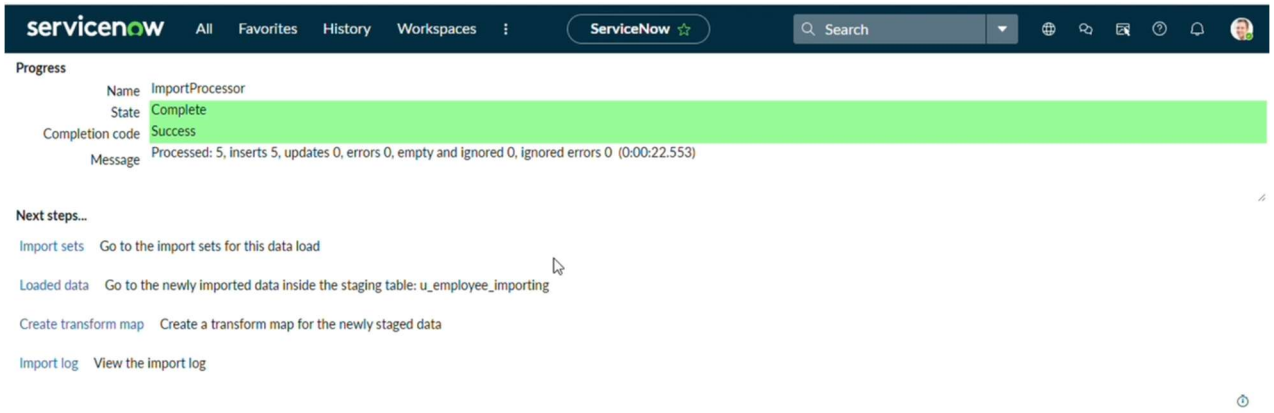


Milestone 2: Creation Of importset Table And Transform Map

Activity 1: Create Importset Table

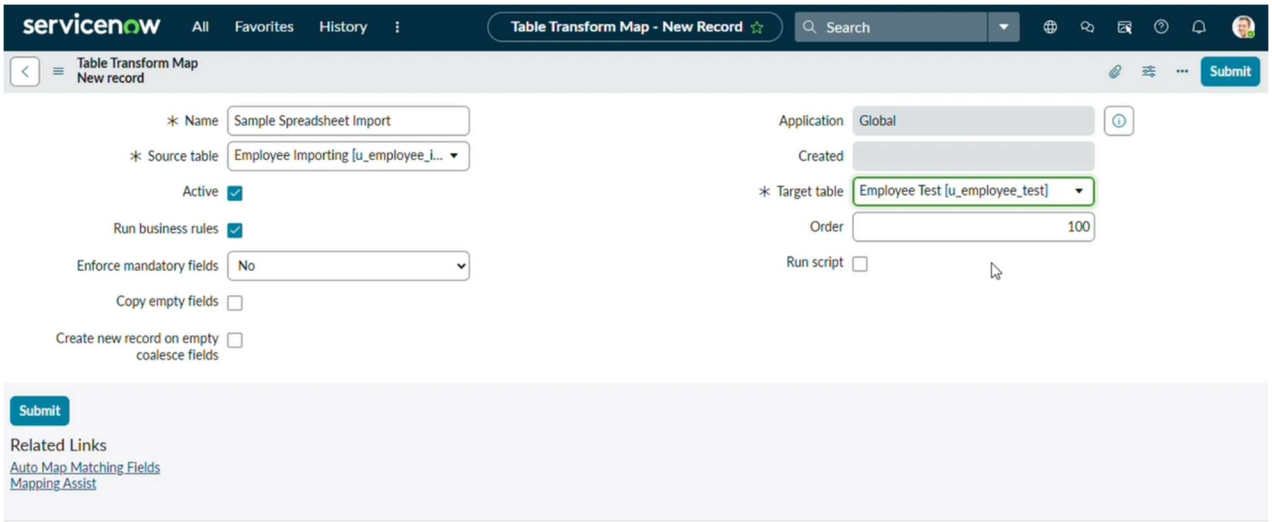
- Go to Application Navigator
- Load Data
- Click on Load Data and create an import table
- Fill details as shown below
- Label: Employee Import
- Name: u_employee_import (auto populated)
- Click on Submit
- Once we click on Submit, we can see the success message as shown below.
- Also we need to make sure to click on Create Transform Map.

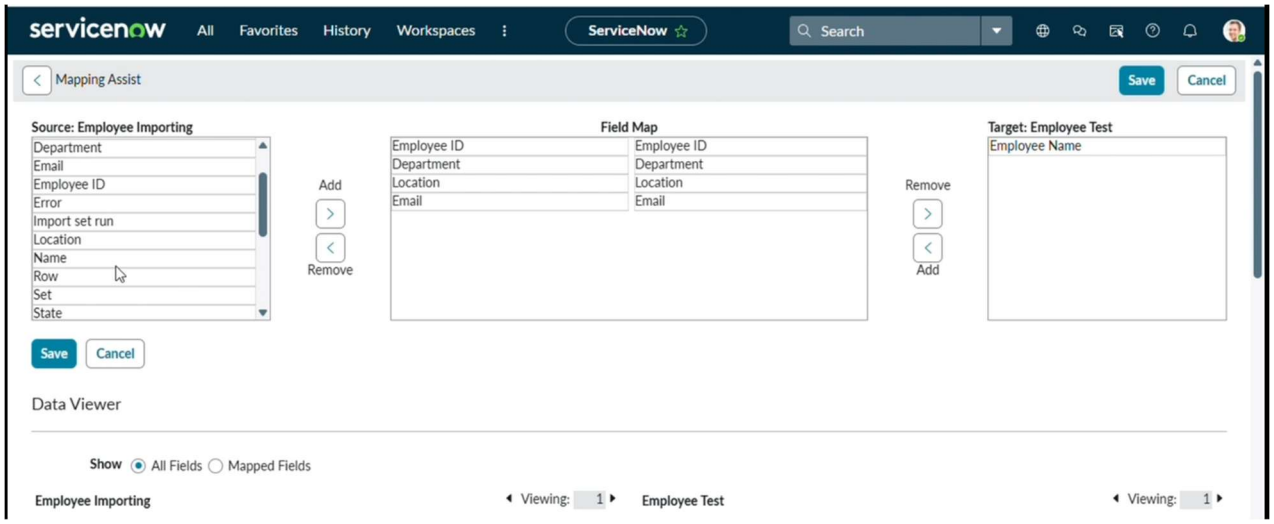


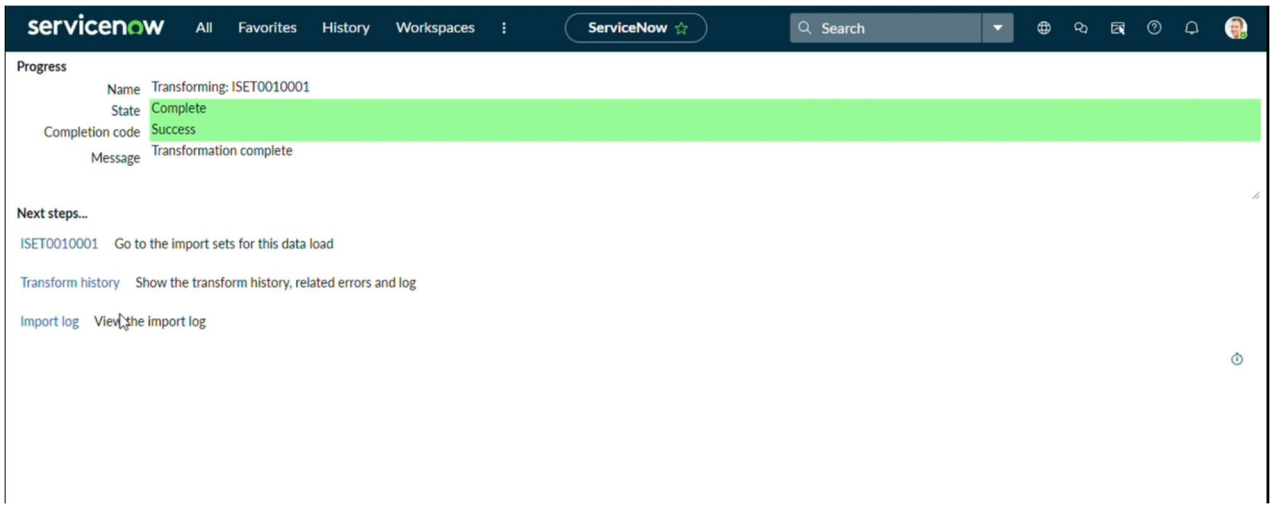


Activity 2: Create Transform Map

- Give inputs as shown below.
- Name: Sample Spreadsheet Import
- Source Table: Employee Import is auto-populated
- Target Table: Employee Test - as we created earlier
- Click on Auto Map Matching fields.
- Click on Save
- Make sure the both Source & Target table fields are mapped as shown below.
- Now click on Transform as shown below
- Now click on Transform link as shown below
- After clicking Transform, make sure you see the success message as shown below.



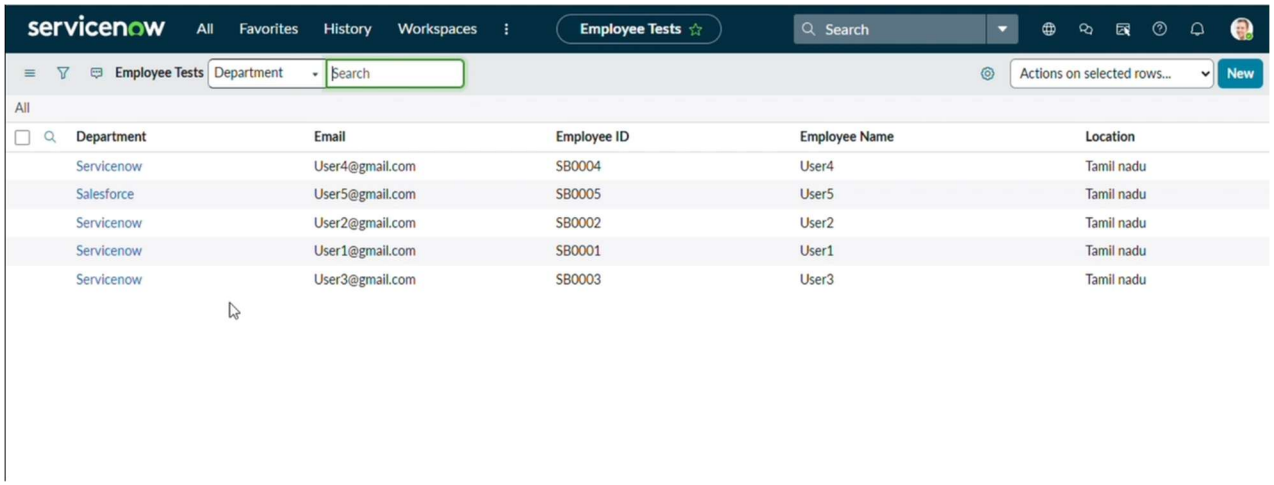




Milestone 3: Transform Data & Validate And Enable Coalesce to Avoid Duplicate Records

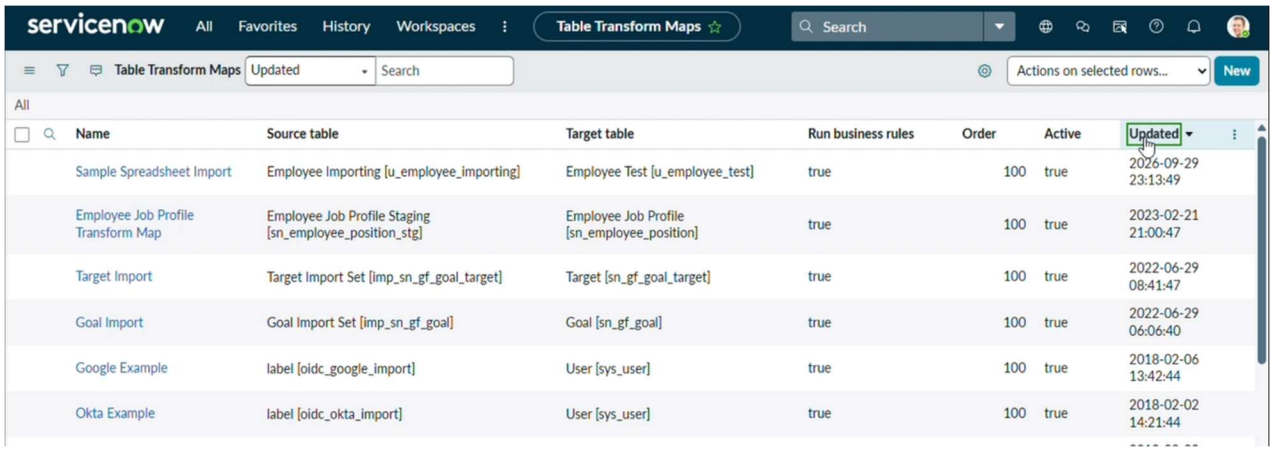
Activity 1: Transform Data & Validate

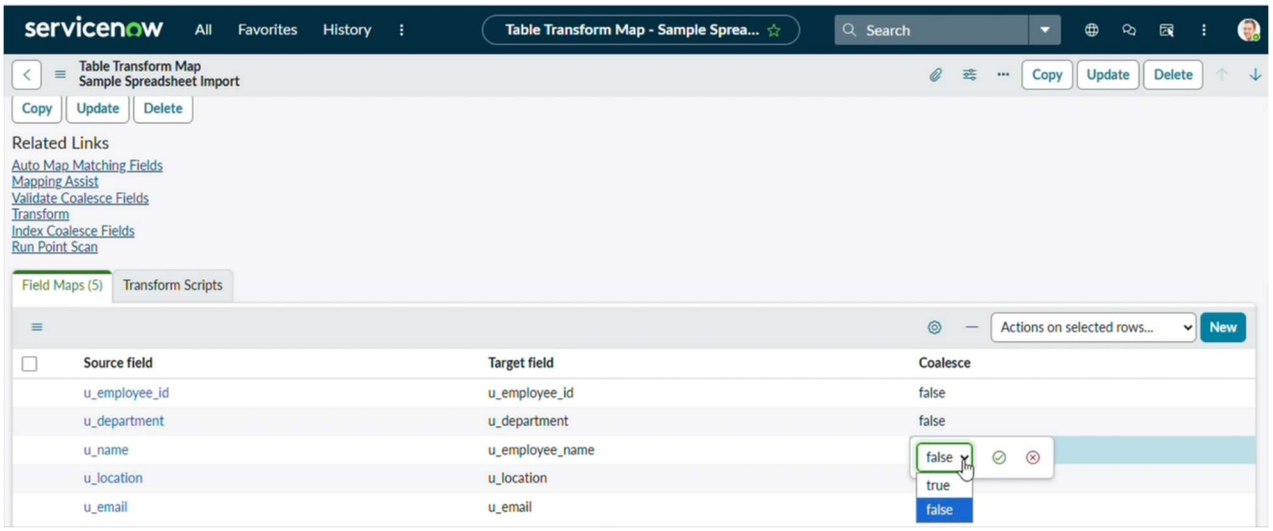
- Open Employee Test Table from application navigator as shown below.
- As our import got succeeded, all the records are created in the Employee Test Table.
- Make sure you rearrange the fields accordingly by using Personalised List Columns
- Click on OK
- Final Result:
- All the records are successfully imported into the ServiceNow system with Import Set, Transform Map.



Activity 2: Enable Coalesce to Avoid Duplicate Records

- Open Transform map Under System import sets On All menus.
- Open the Created Transform map , Name is Sample Spreadsheet Import.
- Under Related list, select the Field maps Section and Make any one field Coalesce True which appears in False.
- Now save the form.
- Then again add some data which is in Excel format.
- Open Transform map Which You are Created
- Go to field mapping
- Go to coalesce field make true for one unique field

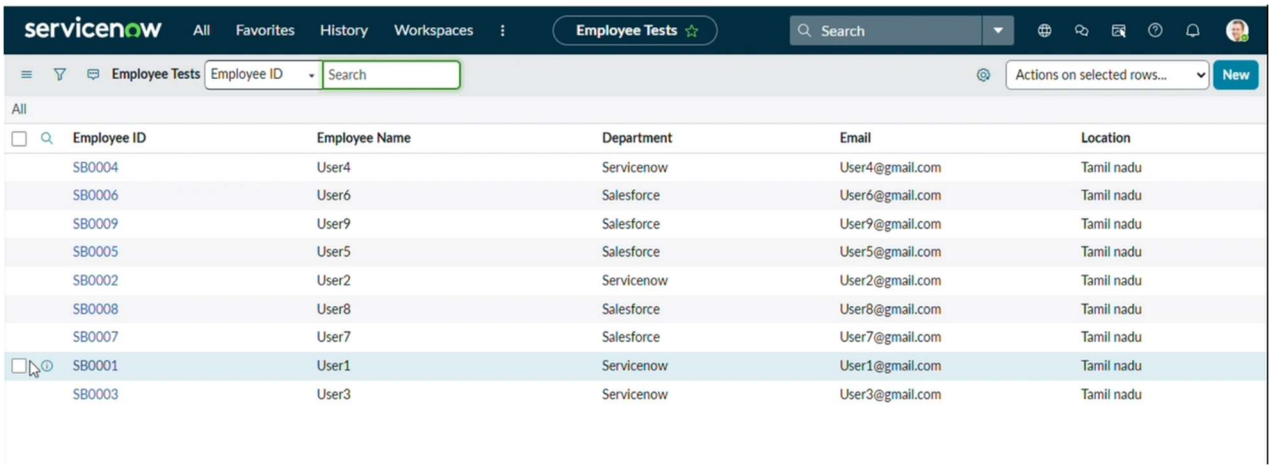




Activity 3: Inserting New Data In Excel Format

- Navigate to All menus.
- Search for Load data Under System Import set Open Load Data.
- Fill the details as
- Import set table as select Existing table {Employee Import}.
- Source of Import is Select File and choose the file from Files.
- Sheet number : 1, Header row : 1.
- Now Click submit.
- Newly Created Excel Data is this.
- It is the previously added data.
- Now, In this new data we added 4 rows of the 2 rows with the same existing data with different Name changed and Email changed.
- That Raviteja Employee Email was Changed to
- [test10@gmail.com → test18@gmail.com].
- And Employee ID with SB-OOO4 name was changed to
- [Ajay kumar → Ajay].
- And two other new Employee ID’s are added.

- Now after submitting the form it appears as State Completed and Success.
- Now Click the Run transform Under the next steps.
- Then it opens Specify Import set and transform map which is previously selected transform map then Click Transform button.
- Now the State is completed and Success.
- Now again click on the Transform history then it shows the newly added data and its status of Inserted and Skipped values out the given data.
- From the above figure : total 4 rows are uploaded out of 2 are Inserted that are two new ID's.
- And 2 rows are updates that are Same ID's with different Name and different Email that we changed.
- Then Open the Employee Tests table and see the rows updated and inserted.
- Now Repeat the Phase 7 with the same steps and same Excel sheet and Submit to check how many rows are updated and Inserted with the same data.
- Then Open the Transform history to check
- Here we can see the total 4 are uploaded.
- On that 0 rows are Inserted and updates 0 and Ignored 4 rows.
- That same sheet we uploaded to the same transform map we created.



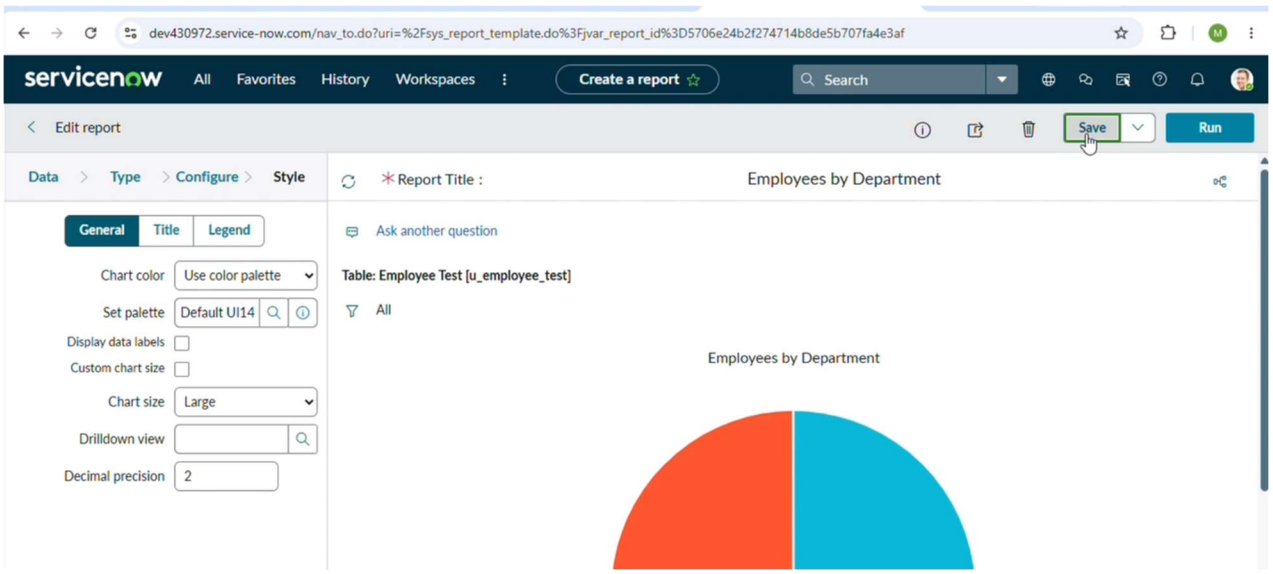
Employee ID	Employee Name	Department	Email	Location
SB0004	User4	Servicenow	User4@gmail.com	Tamil nadu
SB0006	User6	Salesforce	User6@gmail.com	Tamil nadu
SB0009	User9	Salesforce	User9@gmail.com	Tamil nadu
SB0005	User5	Salesforce	User5@gmail.com	Tamil nadu
SB0002	User2	Servicenow	User2@gmail.com	Tamil nadu
SB0008	User8	Salesforce	User8@gmail.com	Tamil nadu
SB0007	User7	Salesforce	User7@gmail.com	Tamil nadu
SB0001	User1	Servicenow	User1@gmail.com	Tamil nadu
SB0003	User3	Servicenow	User3@gmail.com	Tamil nadu

Milestone 4 : Creation Of reports & dashboards

Activity 1: Report 1

- Open All menus, Search for Reports.
- Under Usage and governance Select Reports.
- Click the New button to create a new report.
- Create 3 Reports as follows.
- Report 1: Employees by Department

- In Data
- Report name : Employees by Department.
- Source Type : Table.
- Table : Employee Test, Now click the Next button.
- Now after Clicking next in type Select Pie chart .
- And click the next button
- Now In Configure Select Group by Department Field.
- In the Aggregation Field Select Count.
- Then Click next.
- Now In Style Leave the Chart color by default.
- And Click the Run button to Execute the given data Report.
- And Click save button to Save the Report.



Activity 2: Adding Reports to Dashboard

- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- On the top right corner Click Create Dashboards or
- You can create Directly by using the backend name[PA_DASHBOARDS.FORM].
- Then Click New to Create new Dashboards.
- Only the backend name will open the form if the **ACL** has the **pa_dashboards** access to **Admin Overrides** True only.
- Then only we have access to the form
- After Opening the ACL Search the pa_dashboards on Name Column
- Now Open the **Create Operation** row
- Then Click Enable the True for Admin Overrides and Update the Form.
- Then the creation of dashboards will be possible by using the backend name.
- Now Open the **PA_DASHBOARDS.FORM** create dashboard.

- Dashboard name : **Employee Analytics Dashboards.**
- **Save the Form.**
- **Add Reports to Dashboards.**
- **Navigation Steps:**
- Open the **Previously Created Reports.**
- On All menus → type Reports Under **Usage and Governance** select **reports**.
- Open the **Employees by Department**.
- Select the **Edit Report** and Click the **Share** button then the right top corner Select the **Add to Dashboards** Option.
- Then Choose the Created **dashboard name** and Click **Add** button
- Same Like that do the remaining **2 reports** in the same navigation how we added **Employees by Department.**
- The **Employees by Location** and **Employee List Report** these two reports are also added to the created dashboard.
- After Opening the Report Click the **View Report** Button on the right side of the Context menu.
- After Clicking the View Report button, Now Click the **Share** button and then share options are visible on the right side.
- Now click the **Add to Dashboards**
- Now Add the Dashboard that you created called **Employee Analytics Dashboards.**
- Then Click **Add** button.
- Now the **Report** is added to the **dashboards**.
- Now, Same in the process, add the remaining **two reports** to the **dashboard**.
- From the + icon also we add widgets that our reports from there.
- If we want we can also configure the background color
- Here the three Reports are added successfully by varying **Department , Location and List Report.**
- We can also restrict the **Users,Groups and Roles** From Share Option.

